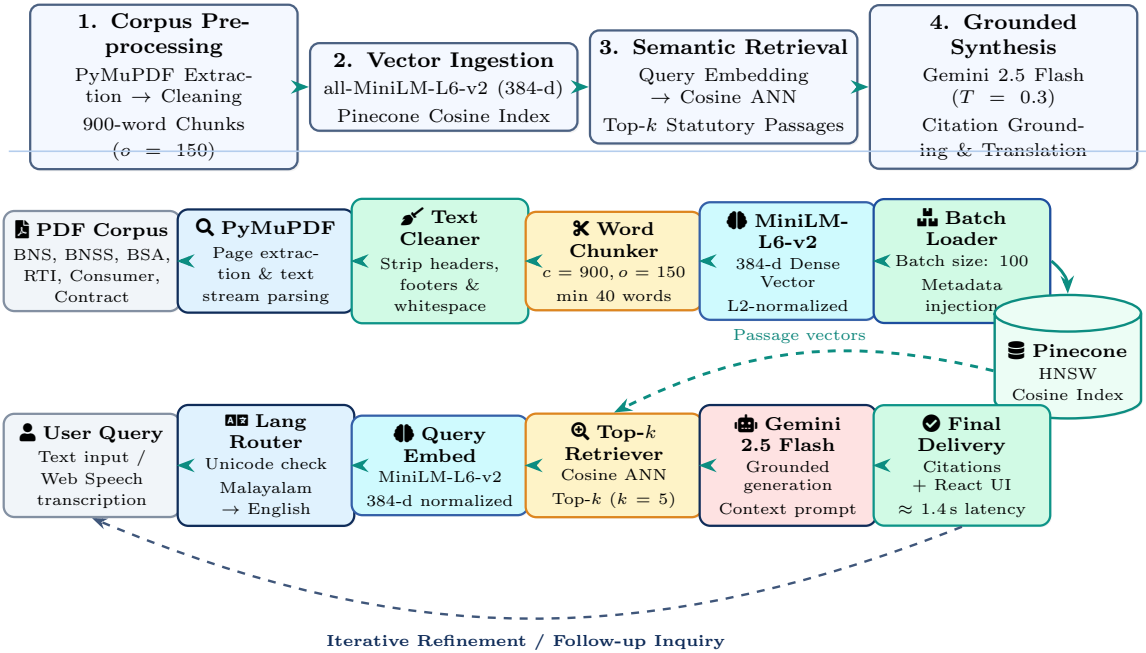

1. Corpus Pre-processing
PyMuPDF Extraction → Cleaning
900-word Chunks ($o = 150$)
2. Vector Ingestion
all-MiniLM-L6-v2 (384-d)
Pinecone Cosine Index
3. Semantic Retrieval
Query Embedding → Cosine ANN
Top-k Statutory Passages
4. Grounded Synthesis
Gemini 2.5 Flash ($T = 0.3$)
Citation Grounding & Translation
PDF Corpus
BNS, BNSS, BSA, RTI, Consumer, Contract
PyMuPDF
Page extraction & text stream parsing
Text Cleaner
Strip headers, footers & whitespace
Word Chunker
$c = 900, o = 150$
min 40 words
MiniLM-L6-v2
384-d Dense Vector
L2-normalized
Batch Loader
Batch size: 100
Metadata injection
Pinecone
HNSW
Cosine Index
Passage vectors
User Query
Text input / Web Speech transcription
Lang Router
Unicode check
Malayalam → English
Query Embed
MiniLM-L6-v2
384-d normalized
Top-k Retriever
Cosine ANN
Top-k ($k = 5$)
Gemini 2.5 Flash
Grounded generation
Context prompt
Final Delivery
Citations + React UI
≈ 1.4 s latency
Iterative Refinement / Follow-up Inquiry